M M Abid Naziri

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About Me

PhD Researcher specializing in **Software Testing** for learning-enabled systems. I develop automated testing techniques for Deep Learning Libraries (**PyTorch**, **TensorFlow**) and **Autonomous Driving Systems** in a simulation environment (**CARLA**). I have reported **140** bugs (**103** confirmed) to **PyTorch** and **TensorFlow** to date [list].

Education

NC State University, PhD in Computer Science (5th year) December 2026 (Expected)

Advisor: Dr. Marcelo d'Amorim
Award: Graduate Merit Award - Summer 2025

Technologies

Languages: Python, R, C#, Delphi, Java, C/C++, JavaScript, TypeScript, Bash

Technologies: PyTorch, TensorFlow, CARLA Simulator, Docker, Github Actions, ASP.NET Core, React JS, Vue JS, TestComplete, DUnit

Experience

Senior Software Engineer, Enosis Solutions – Dhaka, Bangladesh Mar 2019 – Jul 2022

- Developed scalable web applications using Vue JS, TypeScript, and ASP.NET Core, including a project management tool deployed to **270** employees and a Welder Management system
- Developed desktop applications DesignCalcs and ProWrite for pressure vessel design, adding features, fixing bugs, and creating unit/regression tests
- Managed CI/CD pipelines, configured testing servers, and ensured smooth system operations
- Operated in an **Agile** development process in a development team of **7** people
- Mentored new recruits, conducted code/design reviews
- Collaborated with clients and team members via **JIRA**

Research Projects & Publications

Testing Deep Learning Libraries via Neurosymbolic Constraint Learning ICSE 2026
M M Abid Naziri[*], Shinhae Kim[*], Feiran Qin, Saikat Dutta, Marcelo d'Amorim
[*]Equal contribution
Venue: IEEE/ACM International Conference on Software Engineering (ICSE 2026) [pdf]

Misbehavior Forecasting for Focused Autonomous Driving Systems Testing ICSE 2026
M M Abid Naziri, Stefano Carlo Lambertenghi, Andrea Stocco, Marcelo d'Amorim
Venue: IEEE/ACM International Conference on Software Engineering (ICSE 2026) [pdf]

BugsInDLLs: A Database of Reproducible Bugs in Deep Learning Libraries to Enable Systematic Evaluation of Testing Techniques ISSTA 2025
M M Abid Naziri, Aman Kumar Singh, Benjamin Wu, Feiran (Alex) Qin, Saikat Dutta, Marcelo d'Amorim
Venue: International Symposium on Software Testing and Analysis (ISSTA 2025: Tool Demos) [pdf]

Improving Deep Learning Library Testing with Machine Learning AST 2026
Facundo Molina, ***M M Abid Naziri***, Feiran (Alex) Qin, Alessandra Gorla, Marcelo d'Amorim
Venue: ACM/IEEE International Conference on Automation of Software Test (AST 2026) [pdf]

How Effective Is Coverage-Guided Fuzzing to Test Deep Learning Library APIs? ICST 2026
Feiran Qin, ***M M Abid Naziri***, Hengyu Ai, Saikat Dutta, Marcelo d'Amorim
Venue: IEEE International Conference on Software Testing, Verification and Validation (ICST 2026) [pdf]

A Comprehensive Study of Bugs from Deep Learning Library Fuzzers *Submitted*
M M Abid Naziri, Manideepika Reddy Myaka, Lingjun Liu, Saikat Dutta, Marcelo d'Amorim